Honours Project Report
Course Code: CP304

Implementation of Flexible Mechanical Metamaterials Enabling Soft Tactile Sensors with Multiple Sensitivities at Multiple Force Sensing Ranges

Team Members:
Rahul Yadav Entry Number: 2022MEB1334
Sumer Bassi Entry Number: 2022MEB1351

Supervisor: Dr Prabhat K. Agnihotri
Department: Mechanical
Indian Institute Of Technology Ropar

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Contents

1 Abstract

This project focuses on the development of a novel Multi-sensitivity Soft Tactile (MST) sensor based on flexible mechanical metamaterials. The primary objective is to design, model, and simulate a sensor that provides multiple sensitivities across different force sensing ranges, overcoming the traditional limitation of single-sensitivity tactile sensors. The proposed design utilizes a heterogeneous multi-layered structure with tunable stiffness properties, integrated with magnetic-based transduction method for force measurement.

The mechanical metamaterial structure consists of collapsible layers that exhibit step-by-step locking behavior, enabling different sensitivity regimes at various force magnitudes. This project involves CAD design, mathematical modeling, and COMSOL Multiphysics simulation of the metamaterial structure, aiming to demonstrate the feasibility of creating tactile sensors with programmable sensitivity characteristics for applications in robotics, prosthetics, and human-machine interaction.

2 Introduction

Tactile sensing is a fundamental requirement for advanced robotic systems, prosthetic devices, and human-machine interfaces. Traditional tactile sensors are typically designed with fixed sensitivity and operate within a predetermined force range, limiting their versatility in real-world applications. The need for adaptive tactile sensing capabilities has grown significantly with the advancement of soft robotics and the demand for more sophisticated human-robot interactions.

Modern applications such as robotic manipulation, texture identification, slip detection, and in-hand manipulation require tactile sensors capable of operating with different sensitivities across multiple force ranges. Current soft tactile sensors face inherent trade-offs between sensitivity and sensing range, where high sensitivity is achieved at narrow force ranges or low sensitivity across wide force ranges.

This project addresses these limitations by developing a novel approach based on flexible mechanical metamaterials that can provide multiple predetermined sensitivities at different force sensing ranges. The approach is inspired by human tactile sensing capabilities, where the skin's multi-layered structure with varying stiffness properties enables sophisticated touch perception.

The current phase of development involves designing, modeling, and simulating a Multi-sensitivity Soft Tactile (MST) sensor that utilizes mechanical metamaterials as the force transfer medium, integrated with magnetic-based transduction for displacement measurement and force calculation.

3 Mathematical Theory and Formulations

3.1 Fundamental Sensitivity Definition

Based on the original research by Mohammadi et al. (2021), the core mathematical foundation of the MST sensor begins with the definition of sensitivity in terms of magnetic field variation with respect to applied force. The sensitivity S is mathematically defined as:

$$S = \frac{\Delta B / B_0}{\Delta F} \quad [N^{-1}] \tag{1}$$

where:

- S = Sensitivity of the sensor (inverse Newtons)
- ΔB = Change in magnetic field magnitude (Tesla)
- B_0 = Initial magnetic field without applied force (Tesla)
- ΔF = Change in applied force (Newtons)

This fundamental equation establishes the relationship between the measurable magnetic field changes and the applied forces, forming the basis for all subsequent calculations.

3.2 Multi-Layer Mechanical Model

3.2.1 Spring Model Analogy

The metamaterial structure can be mathematically modeled as a system of springs in series, where each unit cell i in block j acts as a spring with stiffness k_{ji}. For a single block with i layers, the force-displacement relationship for each unit cell follows Hooke's law:

$$F_i = k_i \cdot d_i \tag{2}$$

where:

- F_i = Force acting on the i^{th} unit cell (N)
- k_i = Stiffness of the i^{th} unit cell (N/mm)
- d_i = Displacement of the i^{th} unit cell (mm)

3.2.2 Stiffness Hierarchy Condition

For proper step-by-step locking behavior, the unit cells must be arranged in ascending order of stiffness:

$$k_{j1} \ll k_{j2} \ll \cdots \ll k_{ji} \tag{3}$$

This condition ensures that the softest layer deforms first, followed by progressively stiffer layers as the applied force increases.

3.3 Layer-Specific Sensitivity Calculation

For each individual layer i in the multi-layered structure, the sensitivity can be expressed as:

$$S_i = \frac{(\Delta B / B_0)/\Delta d_i}{k_i} \quad [N^{-1}] \tag{4}$$

This equation demonstrates that the sensitivity of each layer depends on:

- The magnetic field gradient with respect to displacement $(\Delta B/B_0)/\Delta d_i$
- The mechanical stiffness of the unit cell k_i

3.4 Force Sensing Range Determination

The force sensing range for each layer i is mathematically defined as:

$$R_i = d_{max_i} \times k_i \quad [N] \tag{5}$$

where:

- R_i = Force sensing range of the i^{th} layer (N)
- d_{max_i} = Maximum displacement of the i^{th} unit cell before locking (mm)
- k_i = Stiffness of the i^{th} unit cell (N/mm)

3.5 Magnetic Field-Displacement Relationship

3.5.1 General Functional Form

The relationship between magnetic field magnitude and permanent magnet displacement can be represented by a function:

$$B = f(d) \tag{6}$$

where:

- B = Magnetic field magnitude (Tesla)
- d = Displacement of permanent magnet (mm)

The specific functional form depends on the magnetic system configuration and can be determined through simulation or experimental measurement.

3.5.2 Normalized Magnetic Field Variation

The normalized change in magnetic field is calculated as:

$$\frac{\Delta B}{B_0} = \frac{B(d) - B_0}{B_0} \tag{7}$$

The gradient of this relationship provides the displacement sensitivity:

$$\frac{\partial}{\partial d}\left(\frac{\Delta B}{B_0}\right) \tag{8}$$

3.6 Complete Sensitivity Calculation Process

3.6.1 Step-by-Step Calculation Method

The complete process for calculating multi-layer sensitivity involves the following steps:

Step 1: Determine Individual Layer Stiffnesses

Using finite element analysis or experimental testing, determine the stiffness values:

$$k_1 = f(\theta_1, t_1, h_1, D_1, H_1, W_1) \tag{9}$$
$$k_2 = f(\theta_2, t_2, h_2, D_2, H_2, W_2) \tag{10}$$
$$k_3 = f(\theta_3, t_3, h_3, D_3, H_3, W_3) \tag{11}$$

where θ, t, and h are the tuning parameters, and D, H, W are the overall dimensions.

Step 2: Calculate Maximum Displacements

For each layer, determine the maximum displacement before locking occurs:

$$d_{max_i} = f(\text{unit cell geometry}) \tag{12}$$

Step 3: Determine Force Ranges

Calculate the force range for each layer using Equation (5):

$$R_1 = d_{max_1} \times k_1 \tag{13}$$
$$R_2 = d_{max_2} \times k_2 \tag{14}$$
$$R_3 = d_{max_3} \times k_3 \tag{15}$$

Step 4: Calculate Magnetic Field Gradients

For each displacement range, calculate the magnetic field gradient through simulation or measurement:

$$\left(\frac{\Delta B}{B_0}\right)_i = f(\text{magnetic system configuration}) \tag{16}$$

Step 5: Compute Final Sensitivities

Apply Equation (4) to obtain the sensitivity for each layer:

$$S_i = \frac{(\Delta B/B_0)_i/\Delta d_i}{k_i} \tag{17}$$

3.7 Design Parameters from Literature

Based on the example provided in Mohammadi et al. (2021), the following design parameters were used:

Optimized Design Parameters:

$$\theta = (25, 33, 37) \tag{18}$$
$$t = (0.6, 1.0, 1.4) \text{ mm} \tag{19}$$
$$h = (2.0, 2.8, 3.5) \text{ mm} \tag{20}$$

Resulting Performance:

$$\text{Sensitivities: } S = (0.26, 0.08, 0.03) \text{ N}^{-1} \tag{21}$$
$$\text{Force ranges: } R = (0.9, 4.8, 16.2) \text{ N} \tag{22}$$

4 Background / Literature Review

4.1 Tactile Sensing Technologies

Tactile sensing has been extensively researched with various transduction mechanisms, each presenting unique advantages and limitations for multi-range sensing applications.

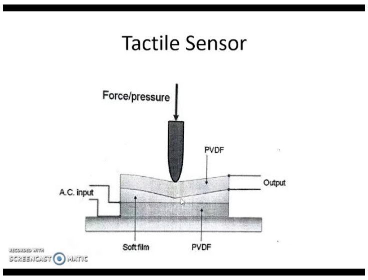


Figure 1: General tactile sensor structure

4.1.1 Resistive and Piezoresistive Sensors

Piezoresistive materials change their electrical resistance under deformation. Good sensitivity but issues like hysteresis and drift [1].

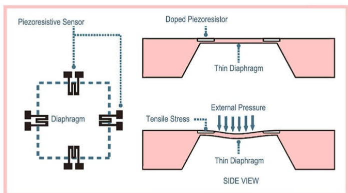


Figure 2: Piezoresistive tactile sensor

4.1.2 Capacitive Tactile Sensors

Measure capacitance change due to deformation; excellent linearity but sensitive to noise and environment [2].

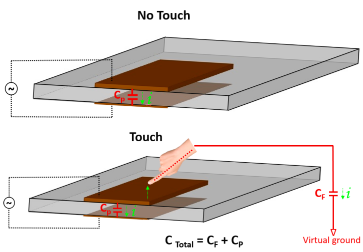


Figure 3: Capacitive tactile sensor

4.1.3 Optical Tactile Sensors

Light-based deformation sensing; high accuracy but bulky and expensive [3].

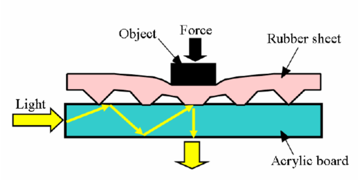


Figure 4: Optical tactile sensor

4.1.4 Magnetic-Based Tactile Sensors

Magnet + magnetometer based sensing; stable, drift-resistant, and suitable for soft robotics [4].

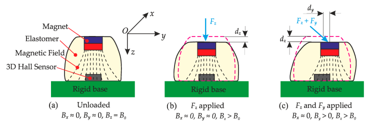


Figure 5: Magnetic tactile sensor

4.2 Mechanical Metamaterials

Mechanical metamaterials are artificially structured materials whose mechanical properties are determined by their geometric structure rather than their constituent materials [7]. These materials offer unprecedented control over mechanical properties through structural design.

Key characteristics of mechanical metamaterials include:

- Programmable stiffness and deformation patterns
- Ability to achieve properties not found in natural materials
- Scalable manufacturing through additive manufacturing
- Tunable mechanical responses through geometric parameters

4.3 Multi-Range Sensing Approaches

Several approaches have been developed to extend the sensing range while maintaining sensitivity:

Hierarchical Structures: Multi-level microstructures that provide different mechanical responses at various force levels [9, 10].

Intrafillable Architectures: Structures with graded stiffness properties that compress progressively under increasing loads [11].

Pressure-Peak Effects: Utilization of structural instabilities to create non-linear force-displacement relationships [12].

However, these approaches typically provide limited control over sensitivity characteristics and cannot achieve arbitrary predetermined sensitivity profiles.

4.4 Research Gap

Despite advances in tactile sensing technology, there remains a significant gap in developing sensors that can provide:

- Multiple arbitrary predetermined sensitivities
- Programmable force sensing ranges
- Simple manufacturing processes
- Integration flexibility for various applications

This project addresses these gaps by developing the MST sensor concept based on flexible mechanical metamaterials.

5 Problem Statement/Aim

5.1 Problem Statement

Existing soft tactile sensors are limited to single predetermined sensitivity and force sensing range, which restricts their applicability in complex real-world scenarios. Applications such as robotic manipulation, prosthetic control, and health monitoring require tactile sensors that can adapt their sensitivity based on the interaction context and force magnitude.

The main challenges include:

1. **Limited Sensitivity Range:** Traditional sensors cannot provide high sensitivity across wide force ranges
2. **Fixed Characteristics:** Sensor sensitivity and range are predetermined during design and cannot be modified
3. **Trade-off Constraints:** Existing designs face inherent trade-offs between sensitivity and sensing range
4. **Manufacturing Complexity:** Multi-sensitivity sensors often require complex fabrication processes

5.2 Aim and Objectives

Primary Aim: To develop and simulate a Multi-sensitivity Soft Tactile (MST) sensor based on flexible mechanical metamaterials that can provide multiple predetermined sensitivities at different force sensing ranges.

Specific Objectives:

1. Develop the mathematical framework and design equations for the metamaterial structure
2. Implement systematic calculation procedures for sensitivity and range determination

3. Design and optimize CAD models using the derived mathematical relationships
4. Develop multi-physics simulation models in COMSOL for validating the theoretical calculations
5. Simulate and analyze the force-displacement characteristics and sensitivity profiles
6. Create and validate the step-by-step locking mechanism through mathematical modeling and FEA
7. Develop the integration approach for the magnetic-based transduction system
8. Create a comprehensive design methodology based on mathematical foundations
9. Prepare for subsequent experimental validation and performance evaluation

6 Methodology

6.1 Integrated Design and Simulation Methodology

This project employs a comprehensive methodology that integrates theoretical design, computational simulation, and systematic validation to develop the MST sensor system. The approach is structured to ensure seamless transition from mathematical concepts to practical implementation.

6.2 Phase 1: Design Requirements and Specifications

6.2.1 Target Performance Based on Literature

Based on the research by Mohammadi et al. (2021), the target specifications are:

Target Sensitivities and Ranges:

- Layer 1: Sensitivity $S_1 = 0.26\ \mathrm{N}^{-1}$, Range $R_1 = 0.9$ N
- Layer 2: Sensitivity $S_2 = 0.08\ \mathrm{N}^{-1}$, Range $R_2 = 4.8$ N
- Layer 3: Sensitivity $S_3 = 0.03\ \mathrm{N}^{-1}$, Range $R_3 = 16.2$ N

6.2.2 Material and Manufacturing Specifications

Based on the original research:

- Material: Thermoplastic Polyurethane (TPU) with Shore hardness A85
- Manufacturing: 3D printing using standard benchtop printers
- Magnet specifications: Disc-shape Neodymium gold coated magnet, diameter 6 mm, thickness 2 mm, remanent magnetization 1.3 T
- Magnetometer: Tri-axis magnetometer (MLX90393, Melexis Inc.) with sensitivity 6211 mT^{-1} and resolution 0.161 μT

6.3 Phase 2: Multi-Physics Simulation Framework

6.3.1 COMSOL Multiphysics Implementation Strategy

The simulation framework employs coupled physics modules to capture the complete sensor behavior:

1. Solid Mechanics Module

- Purpose: Analyze mechanical deformation of metamaterial unit cells under applied forces
- Material Model: Hyperelastic model appropriate for TPU material
- Boundary Conditions: Fixed constraint at base, prescribed displacement or force at top surface

2. Magnetic Fields Module

- Purpose: Calculate magnetic field distribution and variations due to magnet displacement
- Magnet Model: Permanent magnet with specifications matching experimental setup
- Boundary Conditions: Magnetic insulation on outer boundaries

3. Deformed Geometry Module

- Purpose: Couple mechanical deformation with magnetic field changes
- Implementation: Automatic re-meshing based on structural displacement

6.3.2 Simulation Workflow

Step 1: Individual Unit Cell Characterization

1. Create parametric 3D models of unit cells with tuning parameters (θ, t, h)
2. Perform compression simulations under various loading conditions
3. Extract force-displacement data and calculate stiffness values
4. Develop relationships between geometric parameters and stiffness

Step 2: Multi-Layer Assembly Simulation

1. Assemble three-layer system using literature-based parameters
2. Apply progressive loading to demonstrate locking behavior
3. Validate step-by-step compression mechanism
4. Measure individual layer deformation contributions

Step 3: Magnetic System Integration

1. Position permanent magnet within deformable structure

2. Simulate magnetic field distribution for undeformed state
3. Calculate field variations during mechanical deformation
4. Map magnetic field magnitude vs. displacement relationships

Step 4: Coupled Multi-Physics Analysis

1. Execute fully coupled mechanical-magnetic simulation
2. Apply force loads across the expected operating range
3. Extract displacement and magnetic field data simultaneously
4. Calculate sensitivity values using the fundamental equation

6.4 Phase 3: Design Validation and Analysis

6.4.1 Parameter Validation

Validate the simulation framework using the known design parameters:

$$\theta = (25, 33, 37) \tag{23}$$
$$t = (0.6, 1.0, 1.4) \text{ mm} \tag{24}$$
$$h = (2.0, 2.8, 3.5) \text{ mm} \tag{25}$$

6.4.2 Performance Verification

Compare simulation results with the expected performance:

$$\text{Target Sensitivities: } S = (0.26, 0.08, 0.03) \text{ N}^{-1} \tag{26}$$
$$\text{Target Force ranges: } R = (0.9, 4.8, 16.2) \text{ N} \tag{27}$$

7 CAD Design and Modeling

7.1 Unit Cell Design

The unit cell design is based on the mathematical relationships and parameters from the literature. Each unit cell is designed with specific geometric parameters to achieve target stiffness values.

Design Parameters from Literature:

- Depth (D): 20 mm
- Height (H): 5 mm
- Width (W): 30 mm
- Tuning parameters: θ, t, h as specified above

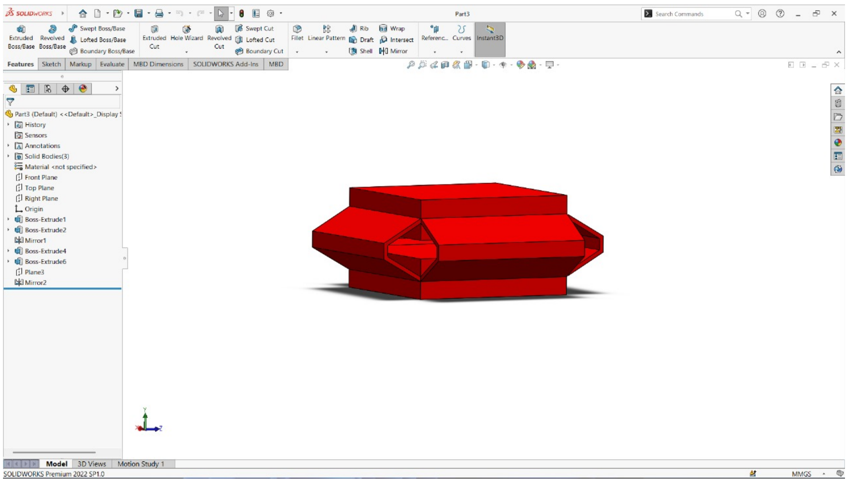


Figure 6: Single unit cell design showing the internal collapsible structure and geometric parameters. The strut architecture enables controlled deformation characteristics under applied loads.

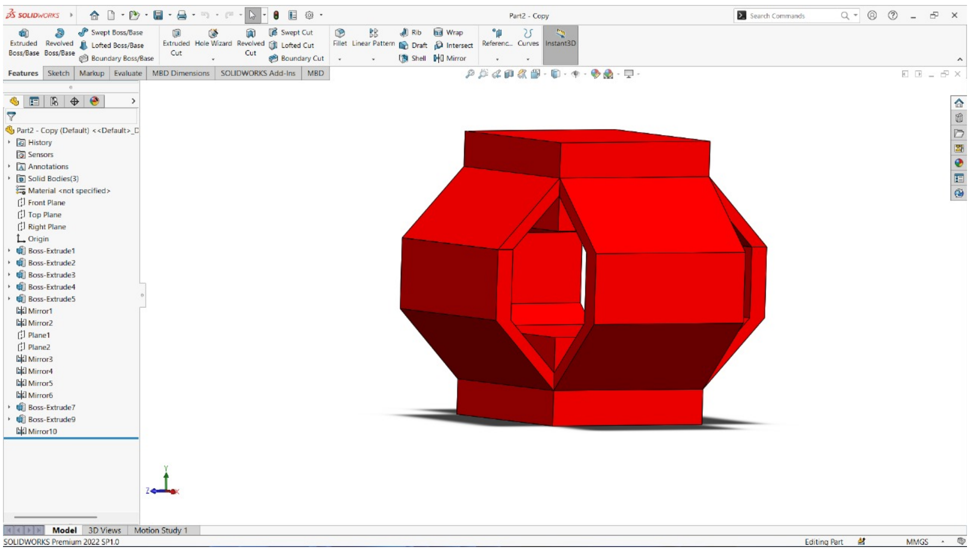


Figure 7: Unit Cell Configurations at Different Geometric Dimension.

7.2 Multi-Layer Assembly

The multi-layer assembly follows the design approach from Mohammadi et al. (2021):

Assembly Configuration:

- Three-layer structure with ascending stiffness values
- Integrated permanent magnet housing in top layer
- Magnetometer housing in base structure
- Proper spacing for mechanical coupling and magnetic field measurement

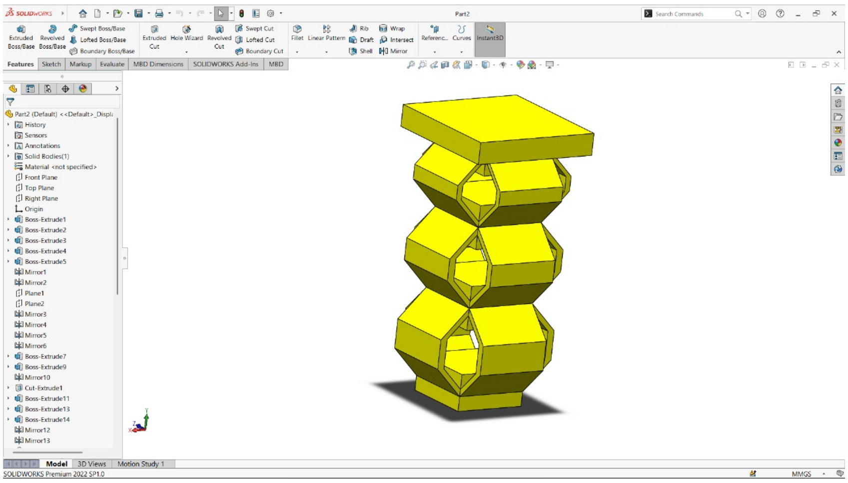


Figure 8: Complete three-layer metamaterial structure showing stacked unit cells with different stiffness properties. The assembly demonstrates the hierarchical arrangement with progressively increasing stiffness from bottom to top layer.

8 COMSOL Multiphysics Simulation Results

This section presents the comprehensive simulation results obtained from COMSOL Multiphysics, covering 2D axisymmetric models, 3D validation models, and fully coupled multiphysics simulations.

8.1 2D Axisymmetric Magnet Simulation

8.1.1 Objective

To simulate the magnetic field strength ($\vec{B}$) from a cylindrical magnet at a point on its central axis. The simulation plots the magnetic field magnitude as a function of the vertical distance (z_disp) between the magnet and the sensor point using a computationally efficient 2D-Axisymmetric model.

8.1.2 Methodology

The simulation was performed using a Stationary study with a Parametric Sweep.

1. **Model Setup:** A 2D-Axisymmetric model was chosen. The *Magnetic Fields (mf)* physics interface was added.

2. **Study:** A *Stationary* study was selected.

3. **Parameters:** Global parameters were defined for magnet radius (`r_mag`), height (`h_mag`), and the initial magnetometer distance (`z_dist`).

4. **Geometry:** The geometry was built in the r-z plane. A rectangle represented the magnet, a larger rectangle represented the surrounding air, and a point at the origin (0, 0) represented the magnetometer.

5. **Materials:** *Air* was applied to the air domain, and *NdFeB N35* (or similar) was applied to the magnet domain.

6. **Physics:** An *Ampère's Law* node was added to the magnet domain, configured with a *Remanent flux density* $\vec{B}_r$ (e.g., $\vec{B}_r = (0, -1.2\ [\mathrm{T}])$).

7. **Study Setup:** A *Parametric Sweep* was added to the *Stationary* study. The sweep was configured to vary the `z_dist` parameter from a large value (e.g., 50 mm) to a small value (e.g., 5 mm).

8. **Data Extraction:** A *1D Plot Group* was created. A *Point Graph* was added, selecting the magnetometer point at (0, 0). The y-axis expression was set to `mf.normB` (magnetic flux density norm) and the x-axis was set to the parameter `z_dist`.

8.1.3 Results

The simulation provides both a 2D visualization of the magnetic field at different distances and the desired 1D plot of field strength versus distance.

(a) Magnetic field at z_disp = 50 mm. (b) Magnetic field at z_disp = 5 mm.

Figure 9: 2D Axisymmetric simulation showing the magnetic flux density (mf.normB) at initial and final magnet-sensor distances. The field is significantly stronger at closer proximity.

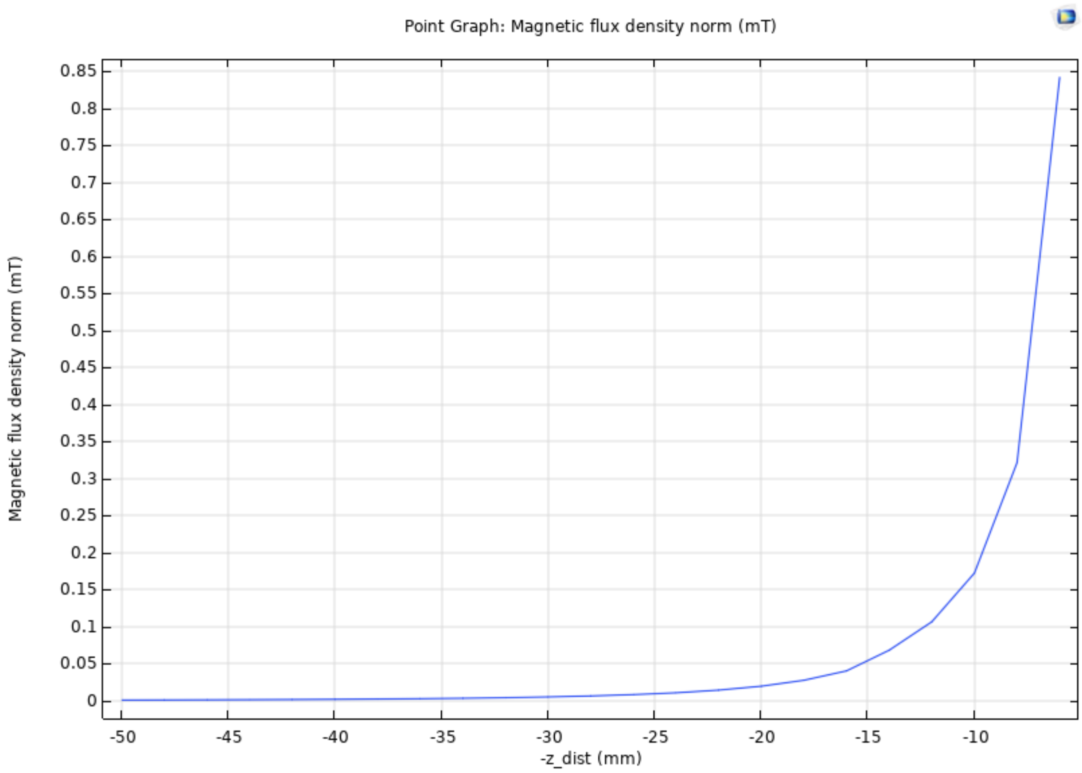


Figure 10: Magnetic field strength at the magnetometer as a function of vertical distance (z_disp) in the 2D model. The field strength increases exponentially as the distance decreases.

8.2 3D Magnet Simulation

8.2.1 Objective

To validate the 2D-Axisymmetric model and to serve as a baseline for the full multiphysics model. This simulation replicates the 2D setup in a full 3D space.

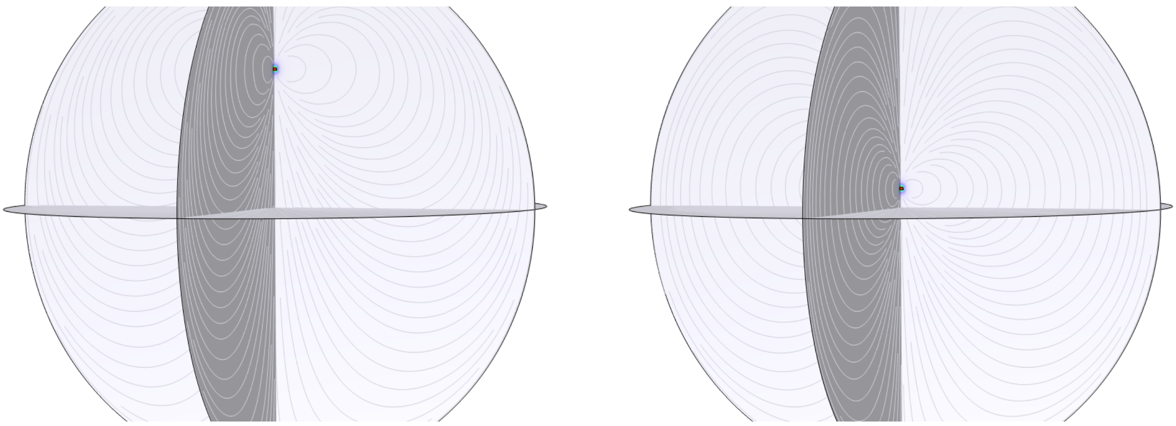
8.2.2 Methodology

The process is nearly identical to the 2D model, but with 3D geometry.

1. **Model Setup:** A 3D model was chosen, adding *Magnetic Fields (mf)* and a *Stationary* study.
2. **Parameters:** Same parameters: `r_mag`, `h_mag`, `z_dist`.
3. **Geometry:** A *Cylinder* was created for the magnet and a large *Sphere* for the air domain. A *Point* at (0, 0, 0) was added for the magnetometer.
4. **Physics:** An *Ampère's Law* node was applied to the cylinder domain with $\vec{B}_r = (0, 0, -1.2\ [\mathrm{T}])$.
5. **Study Setup:** A *Parametric Sweep* was used to vary `z_dist`.
6. **Data Extraction:** A *1D Plot Group* was used to plot `mf.normB` at the magnetometer point against the `z_dist` parameter.

8.2.3 Results

The 3D model results confirm the findings of the 2D model, as expected for a measurement on the central axis.



(a) Magnetic field at z_disp = 50 mm.

(b) Magnetic field at z_disp = 5 mm.

Figure 11: 3D simulation showing the magnetic flux density (mf.normB) on a slice plane at initial and final magnet-sensor distances. The increase in field strength at closer distances is clearly visible.

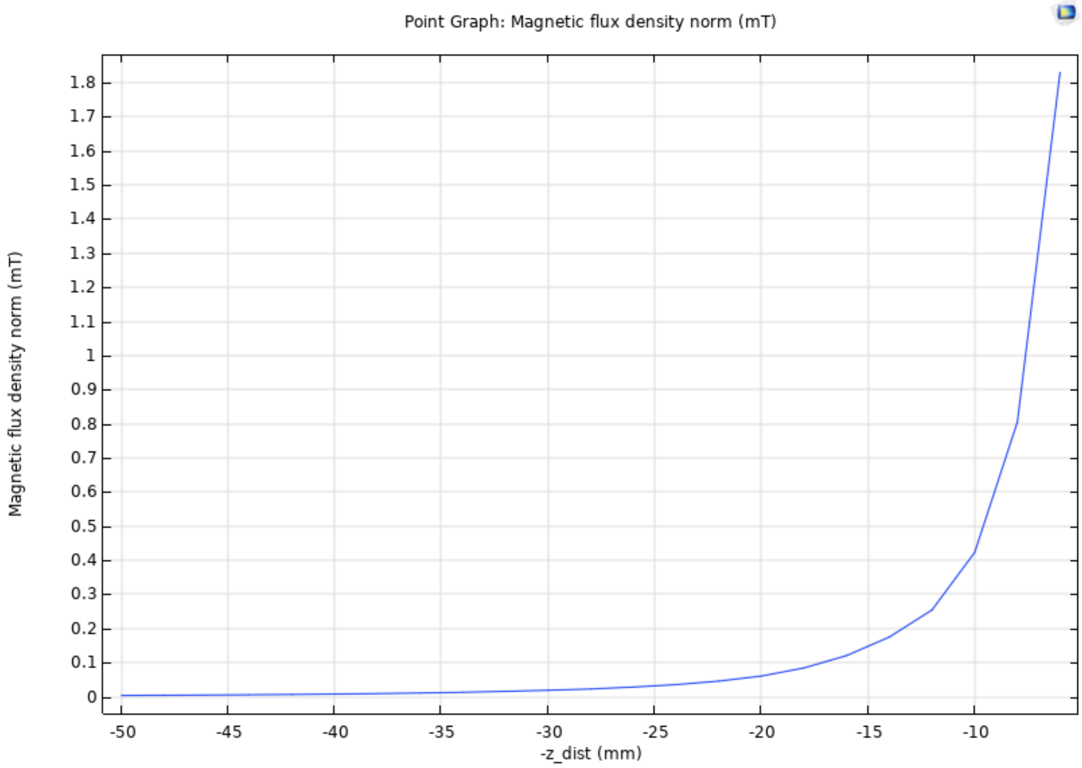


Figure 12: Magnetic field strength vs. vertical distance (z_disp) from the 3D model. This plot should be identical to the 2D-axisymmetric result.

8.3 3D Coupled Multiphysics Simulation

8.3.1 Objective (Cell Structure)

To create a fully coupled model simulating a force-sensing device. An applied force deforms a structure, reducing the distance between an attached magnet and a fixed magnetometer. The goal is to plot the measured magnetic field as a function of displacement, and subsequently, to create a calibration curve to determine the applied force from a change in magnetic field.

8.3.2 Methodology (Cell Structure)

This model couples *Solid Mechanics (solid)*, *Magnetic Fields (mf)*, and *Moving Mesh (ale)*. We use a **prescribed displacement** sweep, which is more stable than a force sweep and directly gives us the plot we need. We then extract the reaction force to find the force-to-field relationship.

1. **Model Setup:** A 3D model was created, adding:
   - *Solid Mechanics (solid)*
   - *Magnetic Fields (mf)*
   - *Moving Mesh (ale)*
2. **Study:** A *Stationary* study with a *Parametric Sweep* was added.
3. **Parameters:** A parameter `y_disp` was created with a value of `0[mm]`. This will be our sweep parameter.
4. **Geometry:** The custom structure was imported. The magnet was partitioned as a separate domain on top. A large air domain (sphere or box) was created around the structure. A point was added to the bottom fixed face for the magnetometer.
5. **Materials:** *Air*, *NdFeB N35* (with mechanical properties like Young's Modulus added), and a structural material (e.g., *Silicone*) were applied to their respective domains.
6. **Physics (Solid Mechanics):**
   - A *Fixed Constraint* was applied to the bottom face of the structure.
   - A *Prescribed Displacement* node was applied to the top face, with the *y*-displacement set to the variable `y_disp`.
7. **Physics (Moving Mesh):**
   - *Prescribed Mesh Displacement (1):* Applied to the solid domains (structure + magnet), with displacement set to `dx=solid.u`, `dy=solid.v`, `dz=solid.w`.
   - *Free Deformation:* Applied to the Air domain.
   - *Prescribed Mesh Displacement (2):* Applied to the outer air boundaries, with displacements set to 0 to keep them fixed.
8. **Physics (Magnetic Fields):**

- *Ampère's Law* was applied to the magnet domain with a remanent flux density (e.g., $\vec{B}_r = (0, -1.2\ [\mathrm{T}], 0)$).
- **Crucial Step:** In the *Discretization* section, the *Spatial frame* was changed to *Mesh (ale)*.

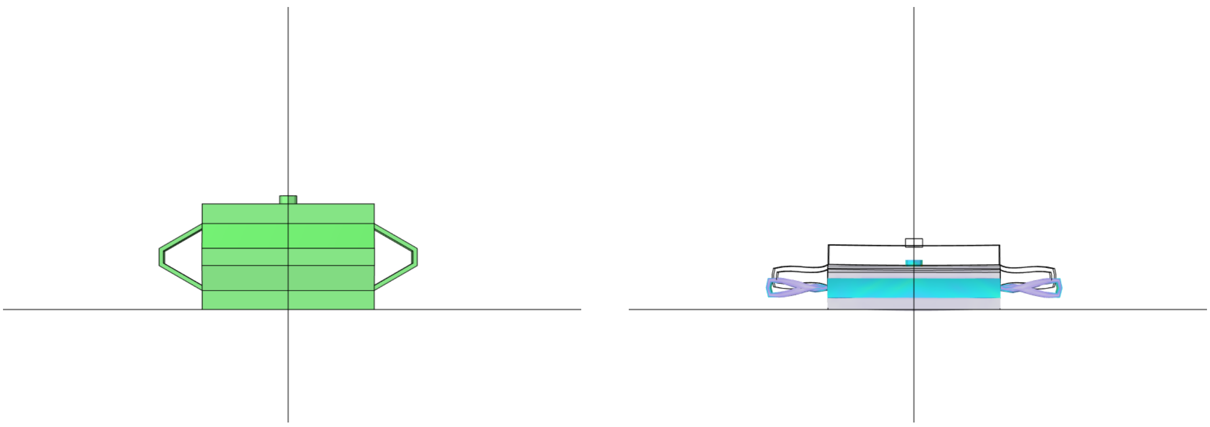
9. **Study Setup:** A *Parametric Sweep* was configured to sweep the `y_disp` parameter from `0[mm]` to `-5[mm]`.

10. **Data Extraction:**

    - 2D/3D plots were generated for stress (`solid.mises`) and magnetic field (`mf.normB`).
    - A 1D Plot Group was created to plot `mf.normB` (at the magnetometer) vs. the parameter `y_disp`.
    - A second 1D Plot Group was created to find the force, plotting the reaction force `solid.RFy` vs. `mf.normB`.

8.4 Results (Cell Structure)

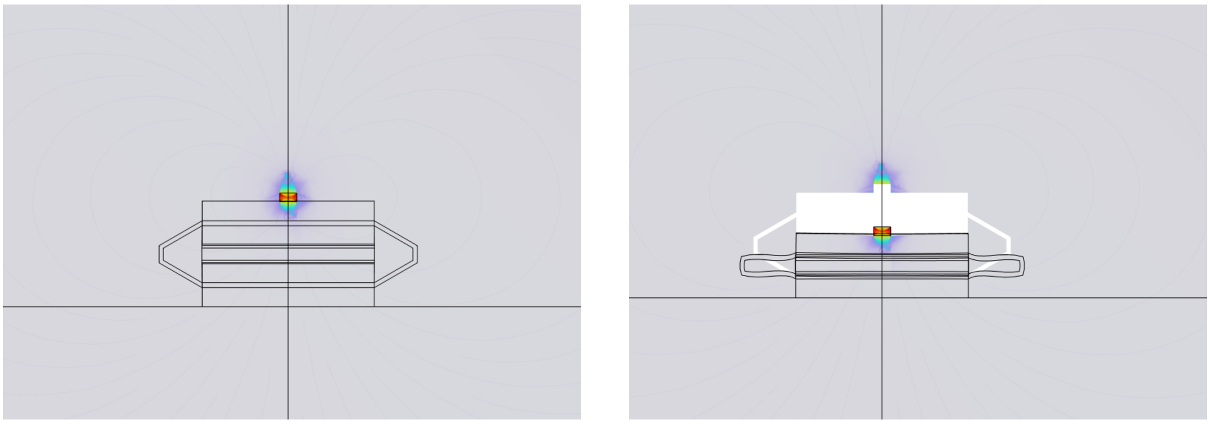
The simulation shows the deformation of the structure and the corresponding change in the magnetic field.



(a) Initial solid deformation (von Mises stress) at y_disp = 0.

(b) Final solid deformation (von Mises stress) at y_disp = -5 mm.

Figure 13: Solid mechanics simulation showing stress distribution at initial and final displacement.



(a) Initial magnetic field at y_disp = 0. (b) Final magnetic field at y_disp = -5 mm.

Figure 14: Magnetic flux density (mf.normB) at initial and final displacement. Note the change in field concentration at the sensor point.

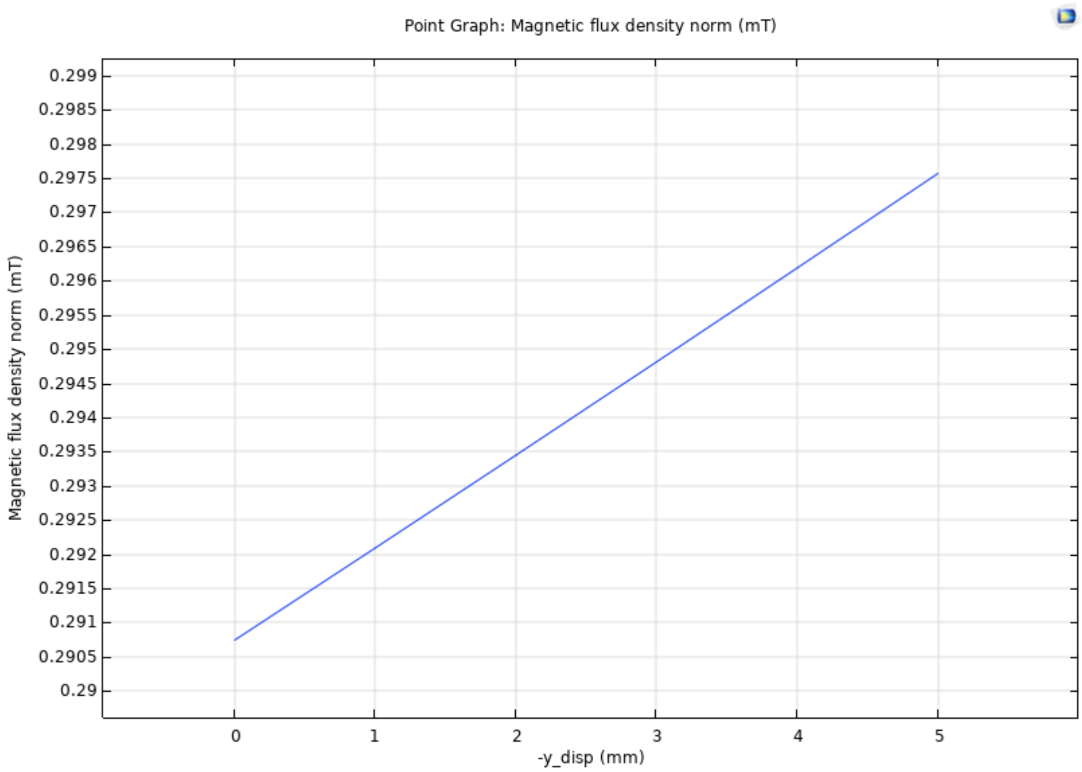
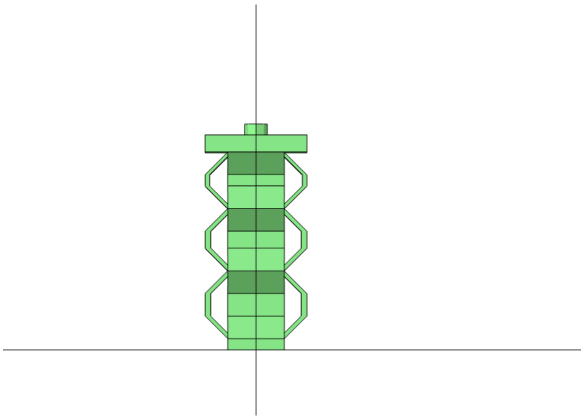


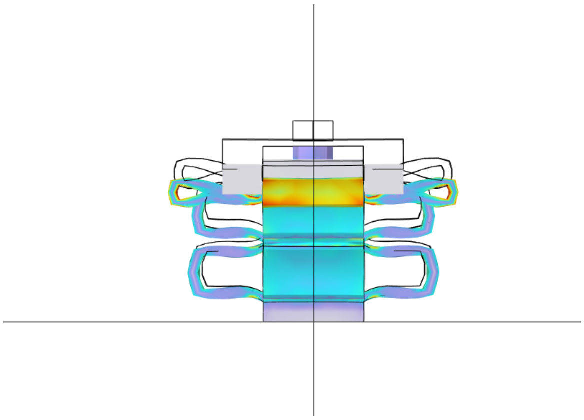
Figure 15: Primary result: Magnetic field strength at the magnetometer as a function of the magnet's vertical displacement (y_disp).

8.5 Results (Block Structure)

The simulation shows the deformation of the structure and the corresponding change in the magnetic field.

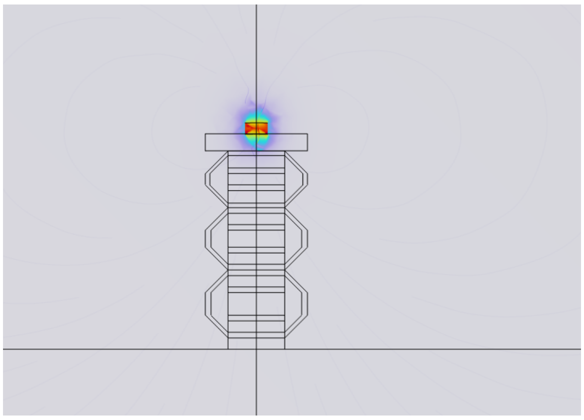


(a) Initial solid deformation (von Mises stress) at y_disp = 0.

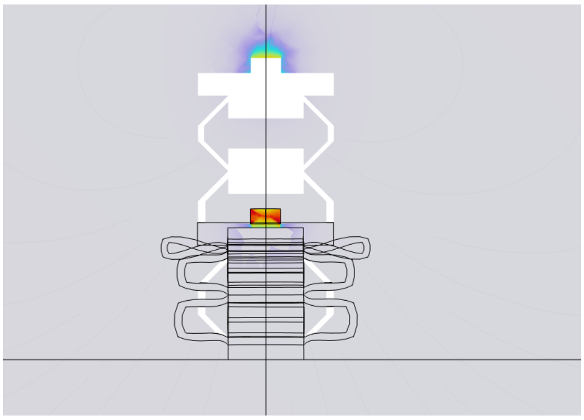


(b) Final solid deformation (von Mises stress) at y_disp = -15 mm.

Figure 16: Solid mechanics simulation showing stress distribution at initial and final displacement.



(a) Initial magnetic field at y_disp = 0.



(b) Final magnetic field at y_disp = -15 mm.

Figure 17: Magnetic flux density (mf.normB) at initial and final displacement. Note the change in field concentration at the sensor point.

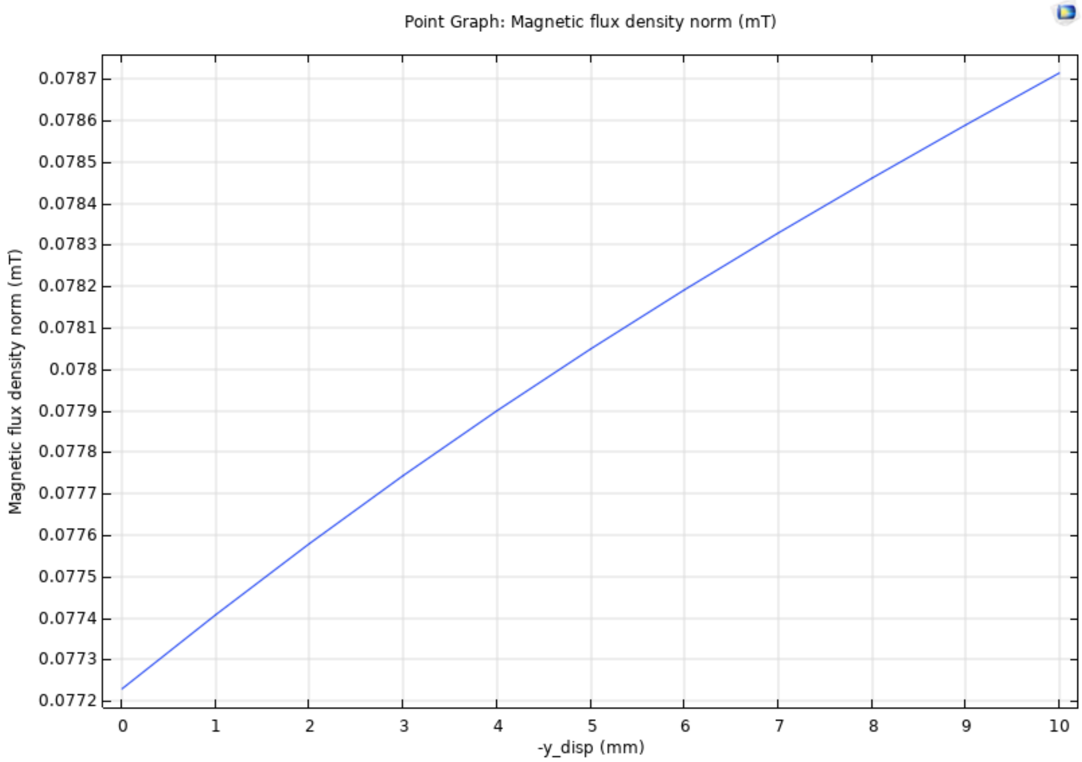


Figure 18: Primary result: Magnetic field strength at the magnetometer as a function of the magnet's vertical displacement (y_disp).

9 Evolution of Methodology and Coupling Challenges

The simulation process underwent significant iterative development to achieve the correct multiphysics coupling and reliable results. This section details the structural iterations, the challenges faced during the multiphysics coupling, and the final successful methodology implemented.

9.1 Challenges Encountered

Integrating the *Solid Mechanics*, *Moving Mesh (ALE)*, and *Magnetic Fields* interfaces introduced several convergence and accuracy-related issues. Multiple strategies were evaluated before achieving a stable and physically consistent solution.

1. **Dependency Error:** During early attempts with a manually configured *Moving Mesh* interface, the solver generated an "Undefined variable `solid.u`" error. This occurred because the ALE interface attempted to access displacement variables before they were computed by the Solid Mechanics module.

2. **Static Magnetic Field ("Ghost Magnet") Issue:** Initial simulations exhibited visible motion of the magnet domain, while the magnetic field distribution remained fixed, resulting in a near-zero response ($\Delta B \approx 0$). The underlying cause was the lack of spatial-frame updating within the Magnetic Fields interface, which occurred because *Geometric Nonlinearity* was initially disabled.

3. **Magnitude Discrepancy in Magnetic Field:** Preliminary field results indicated intensities approximately 10^3 times weaker than expected. This discrepancy was traced to a geometric misalignment introduced during CAD import, which positioned the magnet 10–50 mm away from the intended sensor origin $(0, 0, 0)$.

4. **Desired Magnetic behavior after a distance:** One of the key observation was that for a particular distance our magnetometer point was not sensing the change of magnetic field (because of coupling reason), but after 7 mm movement of the magnet, it senses as required which can be seen in the figure below.

5. **Mesh Discontinuity at the Magnet–Structure Interface:** Applying structural deformation initially caused the magnet to detach or "float" due to the absence of shared mesh nodes at the interface—an issue arising from the use of *Form Assembly* instead of *Form Union*.

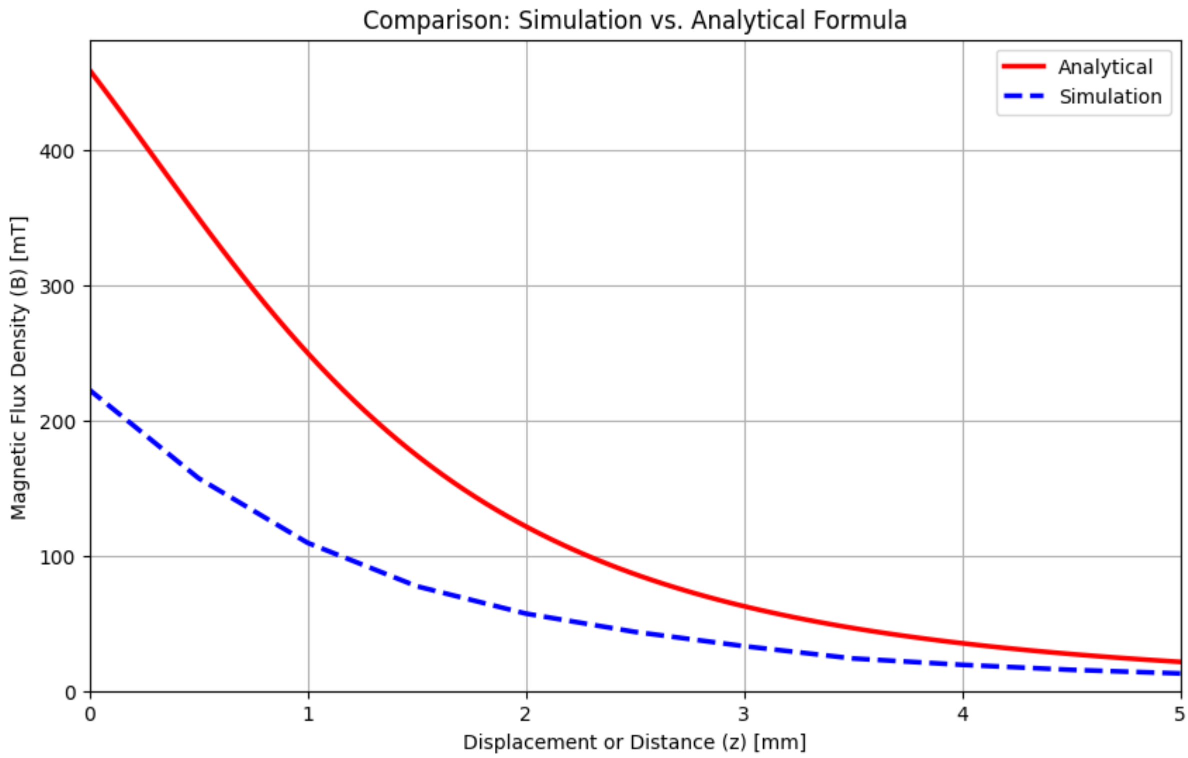


Figure 19: plot comparison between only magnet and magnet with structure

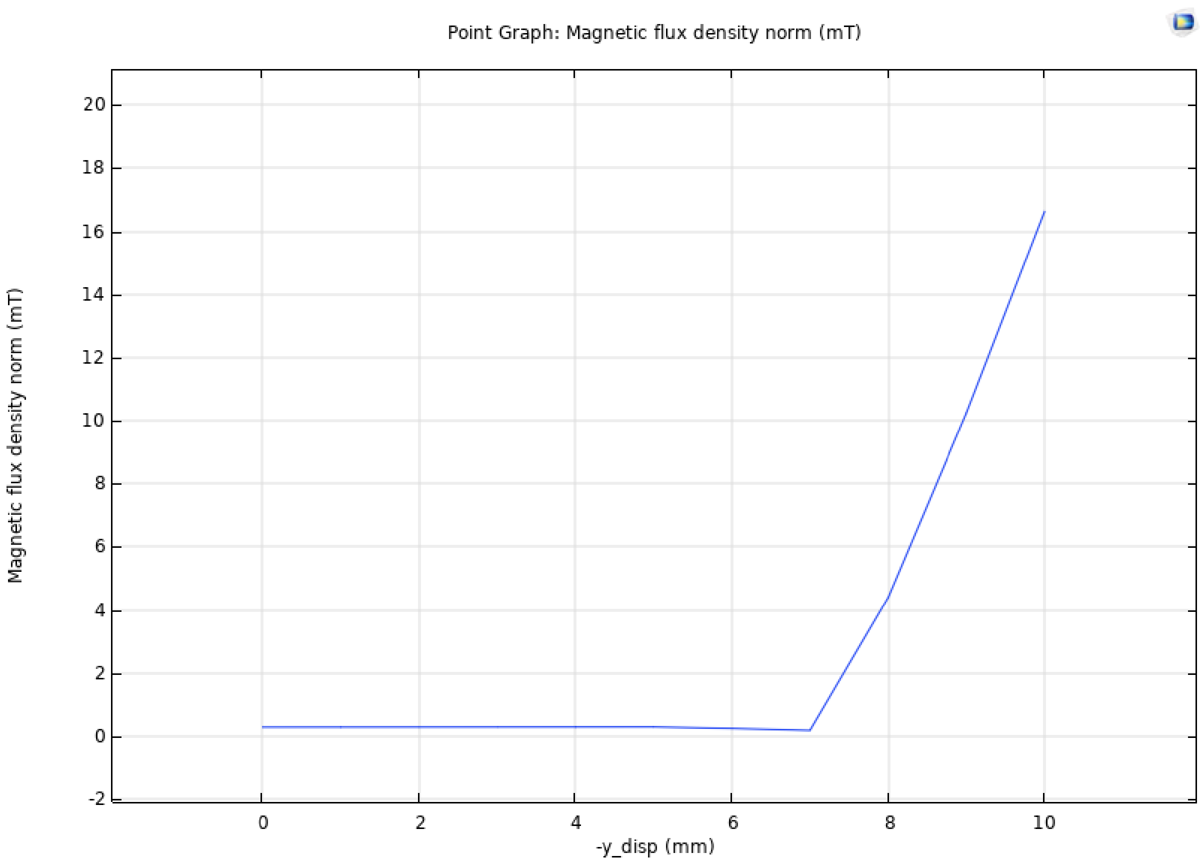
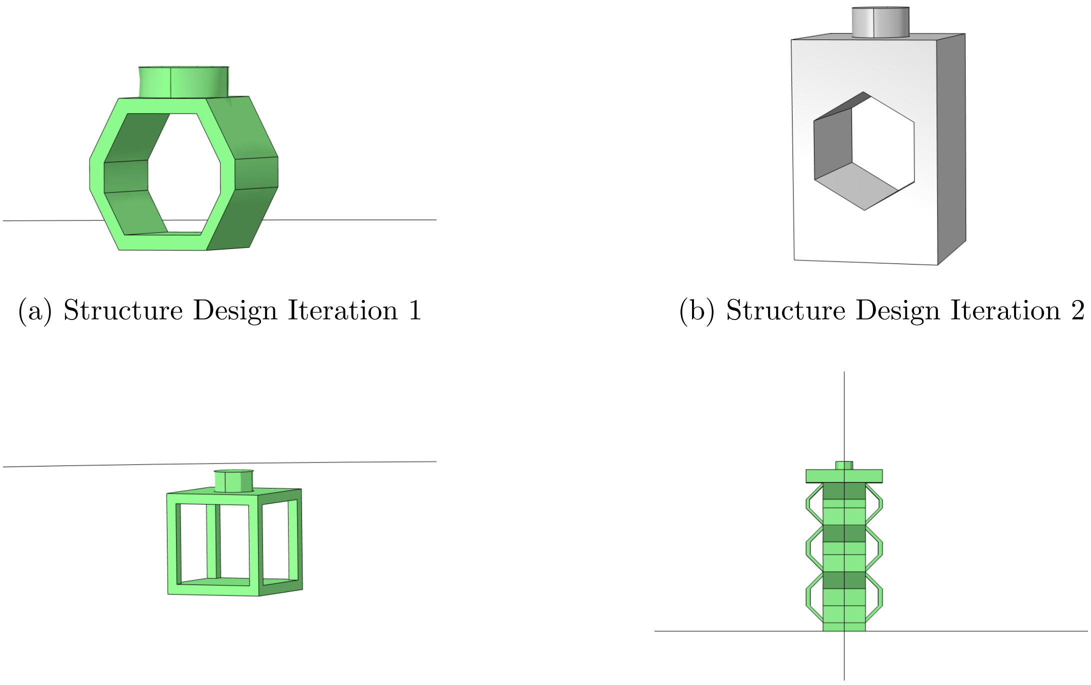


Figure 20: Desired plot after 7 mm

9.2 Structural Designs made to evaluate the problem

To check whether is there any issue with the previous design we made three cell structures in SOLIDWORKS.



(a) Structure Design Iteration 1

(b) Structure Design Iteration 2

(c) Structure Design Iteration 3

(d) Keeping this structure

After checking all the structures we found out that the main problem was the coupling issue of both physics so we kept our old structure for new coupling.

9.3 Coupling Strategies Evaluated

- **Approach A: Manual Coupling** Prescribing deformation in the ALE interface using `solid.u`. *Outcome: Solver failure due to undefined variable dependencies.*
- **Approach B: Manual Frame Adjustment** Attempting to set the Magnetic Fields interface to the *Mesh (ALE)* frame. *Outcome: Configuration instability and persistent static magnetic fields.*
- **Approach C: Multiphysics Coupling Node** Using the built-in *Magnetic–Structure Interaction* node. *Outcome: Improved consistency, though sensitive to boundary condition selection.*

9.4 Final Adopted Methodology

As the whole issue was of coupling of Solid Mechanics and Magnetic field physics, so in our latest approach we are treating the magnet and the base structure as separate identities and the prescribed displacement is applied separately on the top faces of both the identities, and for we are using **Magnetomechanical Forces** as coupling interface which comes under the Multiphysics section, and we are getting satisfactory results. For

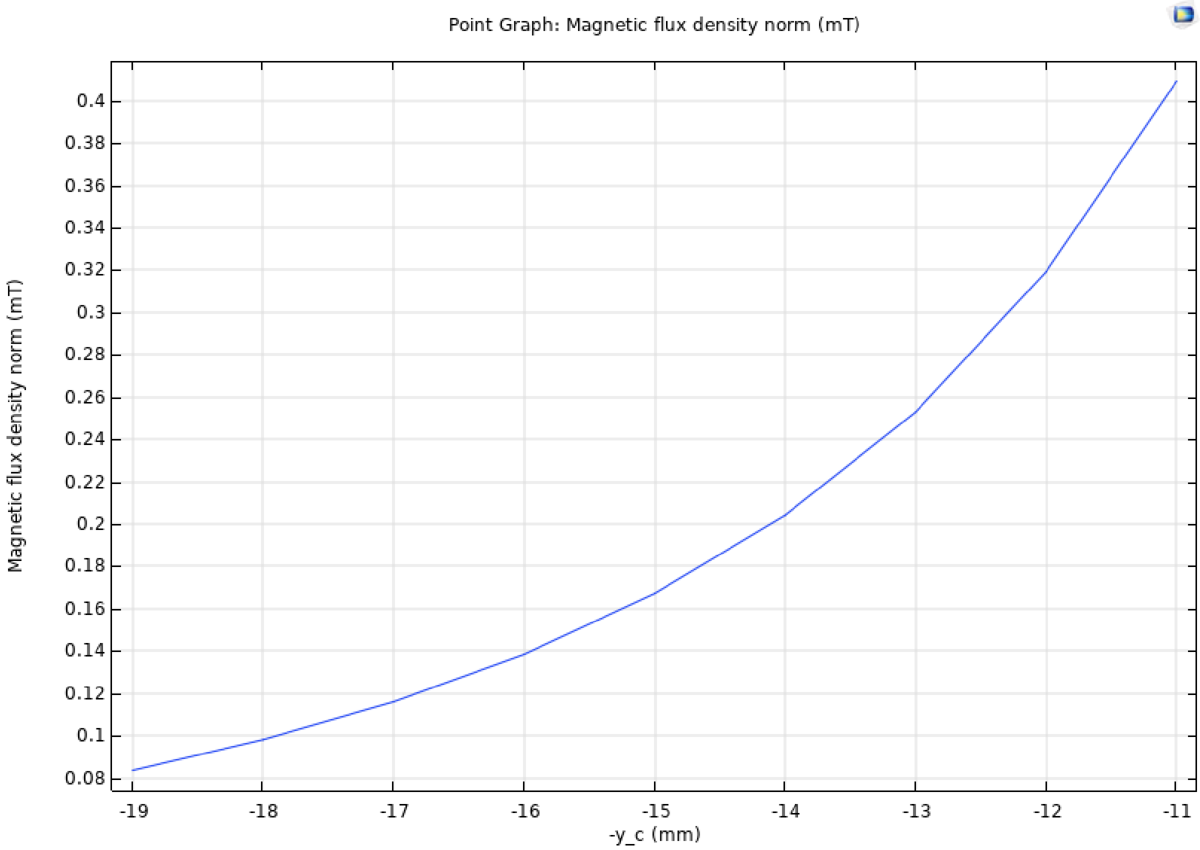


Figure 22: Magnetic flux density vs distance of magnet center from magnetometer

the plot below, the initial magnet position was = 19mm i.e. the top surface of structe and it moved till 11 mm from the magnetometer.

- **Outcome:** This ensured synchronized motion between the magnet and the deforming structure, prevented mesh inversion, and enabled accurate relative motion of the magnetic field source. The resulting magnetic-field predictions aligned closely with reported literature trends, both in magnitude and sensitivity.

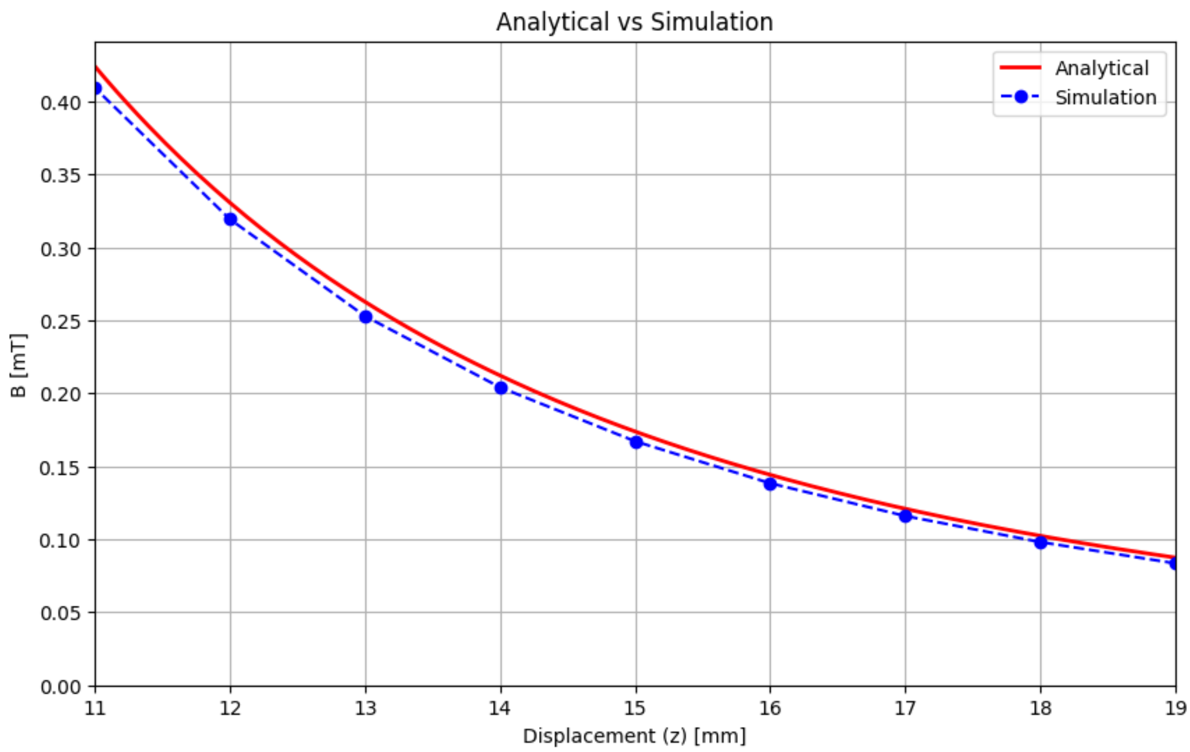


Figure 23: Comparison plot of magnetic flux density for single magnet vs magnet with structure from 19mm(top of structure) to 11mm

9.5 Comparison Plots

```
import numpy as np
import matplotlib.pyplot as plt

Br = 1.3       # Remanent flux density (T)
L = 0.001      # Magnet length/height (m)
R = 0.001      # Magnet radius (m)

z_analytical_m = np.linspace(0.011, 0.019, 200)

term1 = (L + z_analytical_m) / np.sqrt(R**2 + (L + z_analytical_m)**2)
term2 = z_analytical_m / np.sqrt(R**2 + z_analytical_m**2)
B_analytical_T = (Br / 2) * (term1 - term2)

x_analytical = z_analytical_m * 1000
y_analytical = B_analytical_T * 1000

x_simulation = np.array([19., 18., 17., 16., 15., 14., 13., 12., 11.])
y_simulation = np.array([0.083487, 0.098086, 0.11595, 0.13837, 0.16716,
                         0.20414, 0.25296, 0.31933, 0.40938])

plt.figure(figsize=(10,6))
plt.plot(x_analytical, y_analytical, 'r-', lw=2.0, label='Analytical')
plt.plot(x_simulation, y_simulation, 'bo--', lw=1.5, label='Simulation')
plt.xlabel('Displacement (z) [mm]')
plt.ylabel('B [mT]')
```

```python
plt.title('Analytical vs Simulation')
plt.legend()
plt.grid(True)
plt.xlim(11, 19)
plt.ylim(bottom=0)
plt.show()
```

10 Future Work

Future work will concentrate on experimentally validating the simulation results and extending the demonstrated methodology toward application-specific implementations. The primary next steps include:

1. **Experimental Validation:**
   - Fabricate physical prototypes of the MST sensor (e.g., via 3D printing using TPU).
   - Conduct controlled laboratory tests to measure force-displacement and magnetic-field responses.
   - Compare experimental data with simulation predictions and refine material models and boundary conditions.
2. **Comprehensive Performance Characterization:**
   - Quantify sensitivity, linearity, hysteresis, dynamic range, and repeatability for each sensitivity regime.
   - Evaluate temperature and environmental robustness, and characterize sensor drift over time.
3. **Design Optimization:**
   - Perform multi-objective optimization of geometric and material parameters to improve sensitivity–range trade-offs.
   - Explore alternative metamaterial architectures and graded/staged stiffness configurations to achieve tailored sensitivity profiles.
4. **Transduction and Integration:**
   - Investigate alternative transduction mechanisms and hybrid sensing (e.g., magnetic combined with capacitive or piezoresistive elements).
   - Develop compact magnet and magnetometer housings and integrate readout electronics for real-time sensing.
5. **Application-Specific Prototyping and Testing:**
   - Prototype sensor integration into robotic end-effectors, prosthetic devices, and human-machine interfaces.
   - Perform system-level tests (grip tasks, texture recognition, impact monitoring) to demonstrate practical utility.

6. **Scaling and Manufacturability:**
    - Investigate manufacturing scalability (e.g., batch 3D printing, molding) and tolerances for reproducible performance.
    - Assess cost, durability, and long-term reliability for real-world deployment.

Together, these future directions aim to transition the MST sensor concept from simulation-validated designs to robust, deployable sensing systems with demonstrable benefits in robotics, prosthetics, and safety-critical applications.

11 Conclusion

This project presents a comprehensive approach to developing Multi-sensitivity Soft Tactile (MST) sensors based on flexible mechanical metamaterials. The mathematical framework established from the literature provides a solid foundation for systematic design and optimization of sensors with programmable sensitivity characteristics. The current phase focused on validating the theoretical approach through detailed COMSOL Multiphysics simulations, incorporating design parameters and performance specifications from the literature. The simulation-based validation was crucial for understanding the underlying physics and optimizing design parameters for specific applications. The systematic methodology integrating mathematical modeling, CAD design, and multi-physics simulation provides a robust framework for developing application-specific tactile sensors. The demonstrated multi-sensitivity capability represents a significant advancement in tactile sensing technology with broad applicability across robotics, prosthetics, and human-machine interaction fields.

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